Assignment 5

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Title:
Development of a Travel Booking Website using Node.js, Express.js, and MongoDB

Objectives:
The primary goals of this assignment are:

1. To learn how backend systems work using Node.js.
2. To understand how Express.js helps in building web applications easily.
3. To integrate a NoSQL database (MongoDB) with a web application.
4. To design a functional travel booking platform.
5. To allow users to explore travel destinations and make bookings online.

Problem Statement:
The aim of this project is to create a dynamic Travel Agency Website using Node.js and Express framework, with MongoDB used as the database.
The system should allow users to:

- Browse different travel destinations
- Select a preferred travel package
- Enter booking details through a form
- Submit booking information

The application should store all booking data in the database and provide a smooth and interactive user experience.

Outcomes:
After completing this assignment, the following outcomes are achieved:

1. Understanding of server-side programming using Node.js.
2. Ability to create web applications using Express.js.
3. Knowledge of connecting applications with MongoDB.
4. Experience in building a real-time booking system.
5. Understanding the interaction between frontend and backend.

Hardware and Software Requirements
Hardware Requirements

- Computer or Laptop
- Minimum 4GB RAM
- Stable Internet Connection

Software Requirements

- Operating System (Windows/Linux/Ubuntu)
- Node.js installed
- MongoDB database
- Code Editor (VS Code recommended)
- Web Browser (Chrome/Firefox)

Theory

1. Node.js Overview

Node.js is a runtime environment that allows JavaScript to run on the server side. It is built on Chrome's V8 engine and follows an event-driven architecture. This makes it efficient for handling multiple requests simultaneously and suitable for scalable applications.

2. Express.js Overview

Express.js is a framework built on top of Node.js that simplifies backend development. It provides features such as routing, middleware handling, and request-response management, making it easier to build APIs and web applications.

3. MongoDB Overview

MongoDB is a NoSQL database that stores data in a flexible, JSON-like format. Unlike traditional databases, it does not use tables and rows, which allows better scalability and faster development for modern applications.

4. Mongoose

Mongoose is a library that helps in interacting with MongoDB using Node.js. It allows developers to define schemas, validate data, and manage database operations more efficiently.

5. Travel Booking System Concept

In this application:

- Users can view available travel packages
- Users can select a destination
- Users can fill out a booking form
- Booking details are stored in the database

This system simulates a real-world travel booking platform.

6. Working of the Application

The overall working process of the system is as follows:

1. The user opens the website in a browser.
2. Available travel packages are displayed.
3. The user selects a desired package.
4. A booking form is filled with user details.
5. The form is submitted to the server.
6. The backend processes the request.
7. Data is stored in the MongoDB database.
8. A confirmation response is sent to the user.

Flowchart

Start
↓
User visits website
↓
Display travel packages
↓
User selects a package
↓
User fills booking form
↓
Submit form
↓
Server processes data
↓
Store in database
↓
Send confirmation
↓
End

Design / Execution Steps

The development process followed these steps:

1. Installed Node.js and MongoDB on the system.
2. Created a project directory.
3. Initialized the project using npm.
4. Installed required libraries such as Express and Mongoose.
5. Developed the backend server using Express.js.
6. Connected the application to MongoDB.
7. Designed schemas for storing travel and booking data.
8. Created frontend pages using HTML and CSS.
9. Developed a booking form for user input.
10. Connected frontend with backend using routes.
11. Tested the application in a browser environment.

Test Cases

Test Case	Description	Expected Result
Server Start	Run backend server	Server runs successfully
Database Connection	Connect to MongoDB	Connection established
View Packages	Open homepage	Packages displayed correctly
Booking Submission	Submit booking form	Data stored in database
User Interaction	Complete booking	Confirmation message shown

Code Snippet

Node.js + Express

```
const express = require("express");
const mongoose = require("mongoose");
const app = express();

const PORT = 3000;

// Middleware
app.use(express.json());
app.use(express.urlencoded({ extended: true }));

// MongoDB Connection
mongoose.connect("mongodb://127.0.0.1:27017/travelDB")
.then(() => console.log("Database Connected"))
.catch(err => console.log(err));

// Routes
app.get("/", (req, res) => {
   res.send("Welcome to Travel Agency");
});

app.listen(PORT, () => {
   console.log(`Server running on port ${PORT}`);
});
```

Travel Package Schema (MongoDB)

```
const mongoose = require("mongoose");

const packageSchema = new mongoose.Schema({
   title: String,
   location: String,
   price: Number,
   duration: Number,
   description: String,
   image: String
});

module.exports = mongoose.model("Package", packageSchema);
```

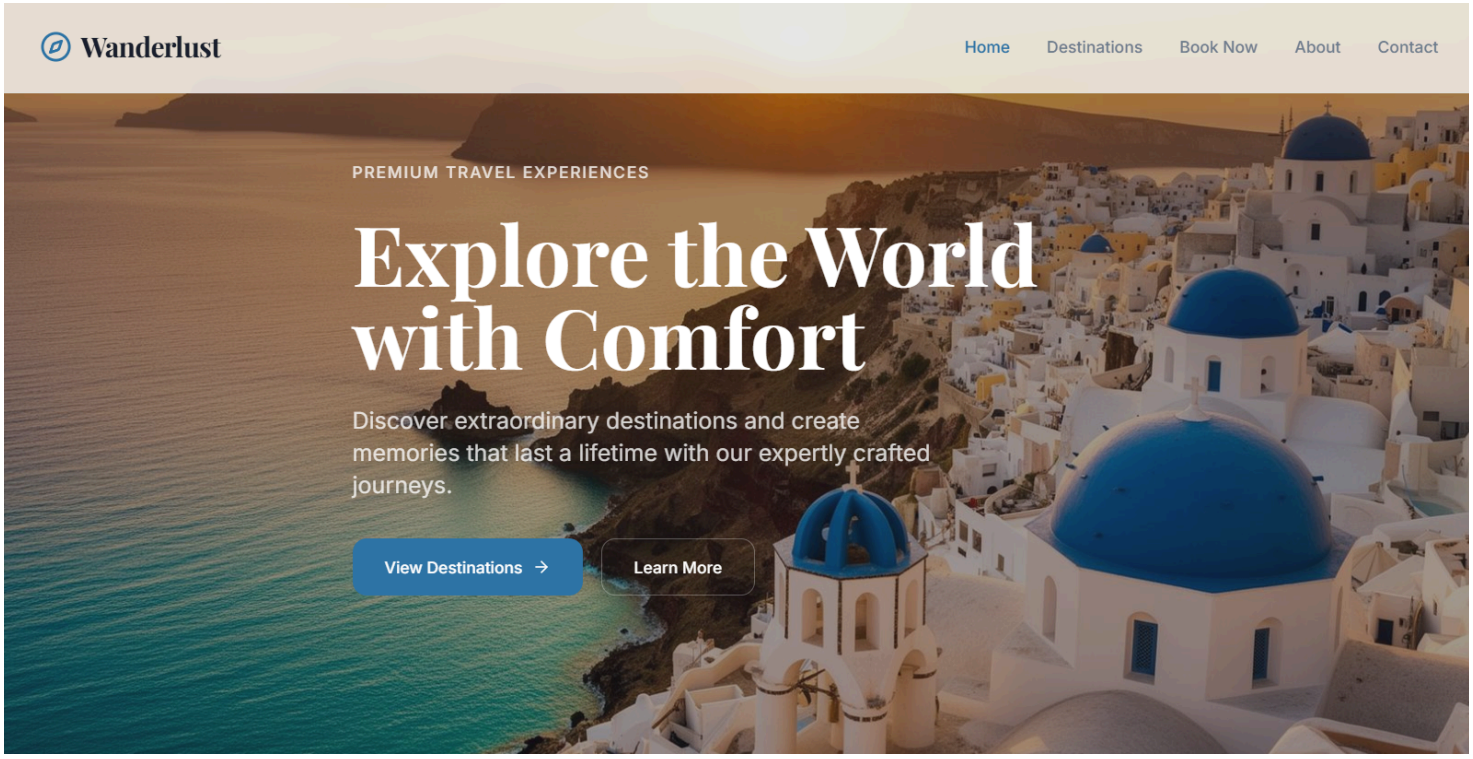
Booking Schema

```
const mongoose = require("mongoose");

const bookingSchema = new mongoose.Schema({
    name: String,
    email: String,
    phone: String,
    travelers: Number,
    travelDate: Date,
    packageId: {
        type: mongoose.Schema.Types.ObjectId,
        ref: "Package"
    }
});

module.exports = mongoose.model("Booking", bookingSchema);
```

Output:



Wanderlust

Home Destinations Book Now About Contact



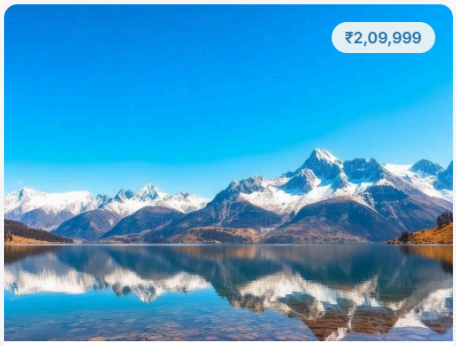
Bali, Indonesia

Bali Serenity Retreat

Discover the magical rice terraces, ancient temples, and pristine beaches of Bali with our curated luxury...

7 Days / 6 Nights ★ 4.8

Book Now



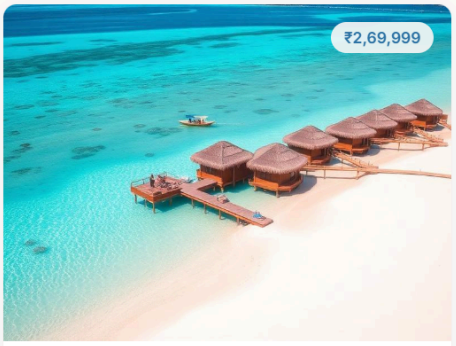
Switzerland

Swiss Alps Adventure

Experience breathtaking mountain vistas, pristine lakes, and charming villages in the heart of the Swiss...

10 Days / 9 Nights ★ 4.9

Book Now



Maldives

Maldives Paradise Escape

Indulge in overwater luxury, crystal-clear waters, and world-class snorkeling in the Maldives.

5 Days / 4 Nights ★ 4.9

Book Now







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Home Destinations Book Now About Contact

Select Package

Japan Cultural Journey — ₹1,59,999

Full Name

John Doe

Email

john@example.com

Phone

+1 555 123 4567

Number of Travelers

1

Travel Date

dd-mm-yyyy

Japan Cultural Journey ₹1,59,999 × 1

Total **₹1,59,999**

Confirm Booking

Conclusion

In this assignment, a complete Travel Agency Website was successfully developed using Node.js, Express.js, and MongoDB. The application enables users to explore travel destinations and book trips through an interactive interface.

The project demonstrates effective integration between frontend and backend technologies, along with proper database handling. It also provides practical experience in building a real-world web application with dynamic functionality.